

```

1  N = 8
2  board = [[0] * N for _ in range(N)]
3
4  def is_safe(row, col):
5      for i in range(col):
6          if board[row][i]:
7              return False
8
9      i = row
10     j = col
11     while i >= 0 and j >= 0:
12         if board[i][j]:
13             return False
14         i -= 1
15         j -= 1
16
17     i = row
18     j = col
19     while i < N and j >= 0:
20         if board[i][j]:
21             return False
22         i += 1
23         j -= 1
24
25     return True
26
27 def solve(col):
28     if col == N:
29         return True

```

Run

```

18         j = col
19         while i < N and j >= 0:
20             if board[i][j]:
21                 return False
22             i += 1
23             j -= 1
24
25         return True
26
27 def solve(col):
28     if col == N:
29         return True
30
31     for row in range(N):
32         if is_safe(row, col):
33             board[row][col] = 1
34
35             if solve(col + 1):
36                 return True
37
38             board[row][col] = 0
39
40     return False
41
42 if solve(0):
43     print("Solution Found:\n")
44     for row in board:
45         print(row)
46 else:

```

Run

Solution Found:

```
[1, 0, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 1, 0]
[0, 0, 0, 0, 1, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 0, 1]
[0, 1, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 1, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 1, 0, 0]
[0, 0, 1, 0, 0, 0, 0, 0]
```

=== Code Execution Successful ===